MIL-L-24532(SH)
17 May 1976
SUPERSEDING
MIL-L-2640D(SHIPS)
30 August 1965
(See 6.4)

MILITARY SPECIFICATION

LIGHT, MARKER, DISTRESS, FLOATING,

FLASHING, AUTOMATIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers automatic floating, flashing, distress electric marker lights.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
L-P-393 - Plastic Molding Material, Polycarbonate, Injection and Extrusion.

MILITARY
MIL-B-18 - Batteries, Dry.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.

STANDARDS

FEDERAL
FED-STD-595 - Colors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Technical paper - Photometer for Measurement of Effective Intensity of Condenser-Discharge Lights (Dated April 1958).

(Application for copies should be addressed to the National Bureau of Standards Section 232.04, Washington, D.C. 20234).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample marker light shall be examined and tested as specified in 4.3 (see 6.3).

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3.2 Materials. Materials shall be as specified herein. However, when no specific material is specified, a material shall be selected which will enable the marker light to meet the performance requirements of this specification.

3.3 Design, construction, and dimensions.

3.3.1 General requirements. Marker light shall consist of the following:

(a) Lens (see 3.3.2.1).
(b) Case (see 3.3.2.2).
(c) Base cone (see 3.3.2.3).
(d) Gaskets (see 3.3.2.4).
(e) Electronic flashing circuit (see 3.3.2.5, 3.3.2.5.1 and 3.3.2.5.2).
(f) Battery (see 3.3.2.6 through 3.3.2.6.3)
(g) Mounting bracket (see 3.3.2.7 and 3.3.2.7.1).

3.3.2 Detail requirements.

3.3.2.1 Lens. The lens shall be made of polycarbonate in accordance with L-P-393. It shall be clear, colorless, with 86 percent minimum light transmission, and ultra-violet (UV) stabilized. The lens shall conform to figure 1.

3.3.2.2 Case. The case shall be made of polycarbonate conforming to L-P-393. The case shall be in accordance with figure 2. Color shall be orange No. 12197 in accordance with FED-STD-595. The case shall be composed of the following three compartments:

(a) Upper compartment shall house the electronic flashing circuit unit.
(b) Lower compartment shall house the battery and battery retaining plug.
(c) Middle compartment shall be void, except for the wire passing through to the end compartments and foam which fills the void to provide flotation if the case should leak. Ends of the middle compartment should be fabricated of 1/8-inch minimum thickness polycarbonate discs cemented to case around their circumference. The wires shall pass through the discs to the upper and lower compartments.

3.3.2.3 Base cone. The base cone shall be polycarbonate conforming to L-P-393 and figure 3. Sufficient ballast shall be included in the base cone to insure that the distress marker light with the base cone attached meets the requirements of 3.5.5 and the tests of 4.5.3 and 4.5.6. Color shall be orange No. 12197 in accordance with FED-STD-595. A 1-inch diameter "O" ring fabricated from 1/8-inch diameter corrosion-resistant steel wire shall be molded into the base cone. The "O" ring shall not separate from the base cone when subjected to a 50-pound pull in any direction.

3.3.2.4 "O" ring gaskets. Two "O" ring gaskets shall be provided, one for the lens and one for the base cone. These "O" ring gaskets shall be commercial neoprene (65-75 durometer hardness) sized to fit in slot of detail A, figure 1. The "O" rings shall be lubricated with a silicone compound, Dow Corning Corp. DC-4 or equal.

3.3.2.5 Electronic flashing circuit unit. Flashing circuit unit shall conform to figure 4. The lamp electronic flashing circuit unit shall be hermetically sealed and able to pass watertightness test (see 4.5.4) as a separate unit. Lamp shall be shock mounted.

3.3.2.5.1 Switch. A dry reed magnetic switch shall be encapsulated in the electronic flashing circuit unit. The switch shall be single pole, single throw, normally open. Contact resistance shall be not more than 0.070 ohms ± 5 percent before and after completion of 1000 operations. Switch shall be open when light marker light is inverted, and closed when light marker light is upright (lens on top).

3.3.2.5.2 Lamp. The lamp shall be a capacitor-discharge xenon flash tube, General Electric FT 106 or equal.

3.3.2.6 Battery. The marker light shall be designed to operate on a type BA-23 battery in accordance with MIL-B-18. The battery shall not be furnished with the marker light.

3.3.2.6.1 A foam plastic spacing cylinder shall be inserted in the battery space during assembly to prevent damage to interior components during shipment and storage.

3.3.2.6.2 Battery retainer plug. Battery retainer plug shall be in accordance with figure 5.

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3.3.2.6.3 Inter-connecting wire between battery and electronic flashing circuit unit shall be of a length so that electronic flashing circuit unit and battery can be removed together without requiring movement or strain on wires. One wire shall be red 18 gage, the other black 18 gage both with closed tooth lugs.

3.3.2.7 Mounting bracket. The mounting bracket shall consist of two spring clips mounted on a strip. With the bracket mounted in its normal position on a vertical supporting surface, the inserted light shall separate from the bracket with a pull of not less than 20 nor more than 50 pounds when such a pull is applied horizontally and directly away from the vertical supporting device.

3.3.2.7.1 Clips shall be made of either spring strip phosphor-bronze, beryllium-copper, or corrosion-resistant steel. Each clip shall be securely fastened to the strip. The strip shall not allow the light to slip out of its bracket due to gravity. All parts of the bracket shall be painted gray. The mounting bracket shall have a neoprene pad secured to it to prevent scratching of the lens at any point where the lens of the light could make contact with the bracket.

3.4 Interchangeability. The lens and electronic flashing circuit unit shall be interchangeable with like items on all distress marker lights conforming to this specificaiton.

3.5 Performance.

3.5.1 Light intensity and flash rate. When fitted with a new battery, the light shall emit intermittent flashes at the rate of 50 ± 10 flashes per minute. The effective candlepower (cp) on a horizonal plane shall be 0.5 cp at 1.49 volts. Light shall be emitted all around the horizon and shall extend downward making an angle of at least 120 degrees with the vertical axis of the unit.

3.5.2 Operation. The light shall operate so that when in the upright position the light shall be "on" and when inverted shall be "off". The light shall operate within a temperature range of minus 30°F to plus 160°F.

3.5.3 Ruggedness. The light complete with battery shall be capable of withstanding the drop and ruggedness tests specified in 4.5.3 and 4.5.6 respectively, without damage or distortion which might impair its ability to meet the requirements of 3.6.1, 3.6.2, and 3.6.4 through 3.6.6.

3.5.4 Watertightness. The marker light shall be watertight and shall show no leakage of water into its interior when tested in accordance with 4.5.4.

3.5.5 Flotation. The marker light, complete with battery, shall be capable of floating for 12 hours in sea water with the lamp center not less than 6 inches above the surface of the water.

3.5.6 Life. The life of the flashing circuit unit shall be a minimum of 500 hours operation.

3.6 designation and marking. The exterior of the case shall be plainly and indelibly marked by molding, engraving, hot stamping, or silk screening with the following information:

U.S. (in 1/2-inch letters)
Light, marker, distress (in 1/4-inch letters)
Specification number
Contract number
Manufacturers name or trademark
National stock number (NSN) 6230-00-782-0643

(and the following in not less than 1/8 inch letters)

IMPORTANT:

1. Check operation once a month.
2. Examine lens for condensation at least ONCE a MONTH.
3. When re-assembling, check "O" rings for nicks and cuts. Replace if necessary, grease with silicone compound, and screw lens and base cone hand tight.
4. Replace battery at least every 12 months.
5. Install battery with terminals facing base cone.
6. Connect red wire to positive terminal and black wire to negative terminal.
7. Secure battery with rubber retaining plug and base cone.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. The sample for first article inspection shall be subjected to the examination of 4.5.1 and the tests of 4.5.2 through 4.5.10.

4.3.1 Test reports. The supplier shall prepare a first article test report in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot. For purposes of sampling, a lot shall consist of not more than 5,000 marker lights offered for delivery at one time.

4.4.1.2 Sampling for examination. A random sample of marker lights shall be selected from each lot in accordance with table I and shall be examined as specified in 4.5.1. Any marker lights in the sample containing one or more defects shall be rejected and, if the number of nonconforming lights in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table I - Sampling for examination.

Number of Marker lights in lot	Number of marker lights in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 to 800	50	2	3
801 to 1300	75	3	4
1301 and over	110	4	5

4.4.1.3 Sampling for tests. A random sample of marker lights shall be selected in accordance with table II and shall be subjected to the tests specified in 4.5.2, 4.5.4, 4.5.5, and 4.5.10. If any sample marker light fails to conform to these tests, the marker light and the lot represented by the marker light shall be rejected.

Table II - Sampling for tests.

Number of marker lights in lot	Number of marker lights in sample
40 and under	3
41 to 110	5
111 to 300	7
301 to 500	10
501 to 800	15
801 and over	25

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4.5 Examination and tests.

4.5.1 Examination. Marker lights shall be subjected to surface examination to determine that design, marking, material, dimensions, and shape are in conformance with the requirements of this specification. Particular care shall be taken to insure that the requirements of 3.4 is verified.

4.5.2 Operation. Marker lights shall be fitted with batteries and operated in various positions to determine compliance with the circuit opening and closing requirements (see 3.5.2).

4.5.3 Drop test. Marker lights, complete with battery, shall be dropped twice from a height of 60 feet into seawater or liquid of equal specific gravity. After each drop, the light shall be examined to determine compliance with 3.5.3.

4.5.4 Watertightness. The marker light, complete with battery, shall be submerged horizontally in salt water (3 percent aqueous solution of NaCl) at a head of 1 foot for a period of 2 hours. At the end of this period, the light shall be removed and the outside wiped dry. The light shall then be disassembled and examined to verify compliance with 3.5.4. The test shall be repeated for flashing mechanism only.

4.5.5 Flotation. Marker light shall be tested to verify compliance with 3.5.5.

4.5.6 Ruggedness. Marker light, complete with battery, shall be cooled to 0°F. It shall then be dropped twice from a height of 3 feet onto a rigidly mounted steel plate. The distance of 3 feet shall be measured from the top of the lens to the impact surface when dropped, the light shall strike on the top center of the lens. The light shall then be examined to determine compliance with 3.5.3.

4.5.7 Light output and flash rate. Effective candlepower of the light shall be determined by the effective intensity method described in the National Bureau of Standards Technical Paper (see 2.2). The flash rate should also be determined to verify conformance with 3.5.1.

4.5.8 Life. The flashing circuit unit shall be connected to a 1.5 direct current source and operated for a total of 500 hours to determine compliance with 3.5.6.

4.5.9 Switch operation. The dry reed magnetic switch shall be operated through 1000 "on-off" cycles to determine compliance to 3.3.2.5.1.

4.5.10 Pull test. The pull required to separate the light from its bracket shall not exceed the limitations specified in 3.3.2.7. The light shall be installed in a rigidly mounted bracket for this test and a perpendicular force applied.

4.5.10.1 The light shall be securely supported and a force of 50 pounds shall be applied to the "O" ring to verify the requirements of 3.3.2.3.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging, packing and marking. Marker lights shall be preserved-packaged level A or C; packed level A, B, or C as specified (see 6.2.1), and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. Marker lights are to be used as life saving aids in conjunction with life rings.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation-packaging and packing required (see 5.1).

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6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9 (n) and which incorporated a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

Specification paragraph	Data requirements	Service	Applicable DID
4.1.1	Inspection system plan	SH	DI-R-4803
4.4.1	First article test reports	SH	UDI-T-23450

(Copies of DID's required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Supersession data. This specification supersedes MIL-L-2648 dated 2 June 1960 and Drawing 815-P-1197249 dated 1 December 1967.

Preparing activity:
Navy - SH
(Project 6230-N216)

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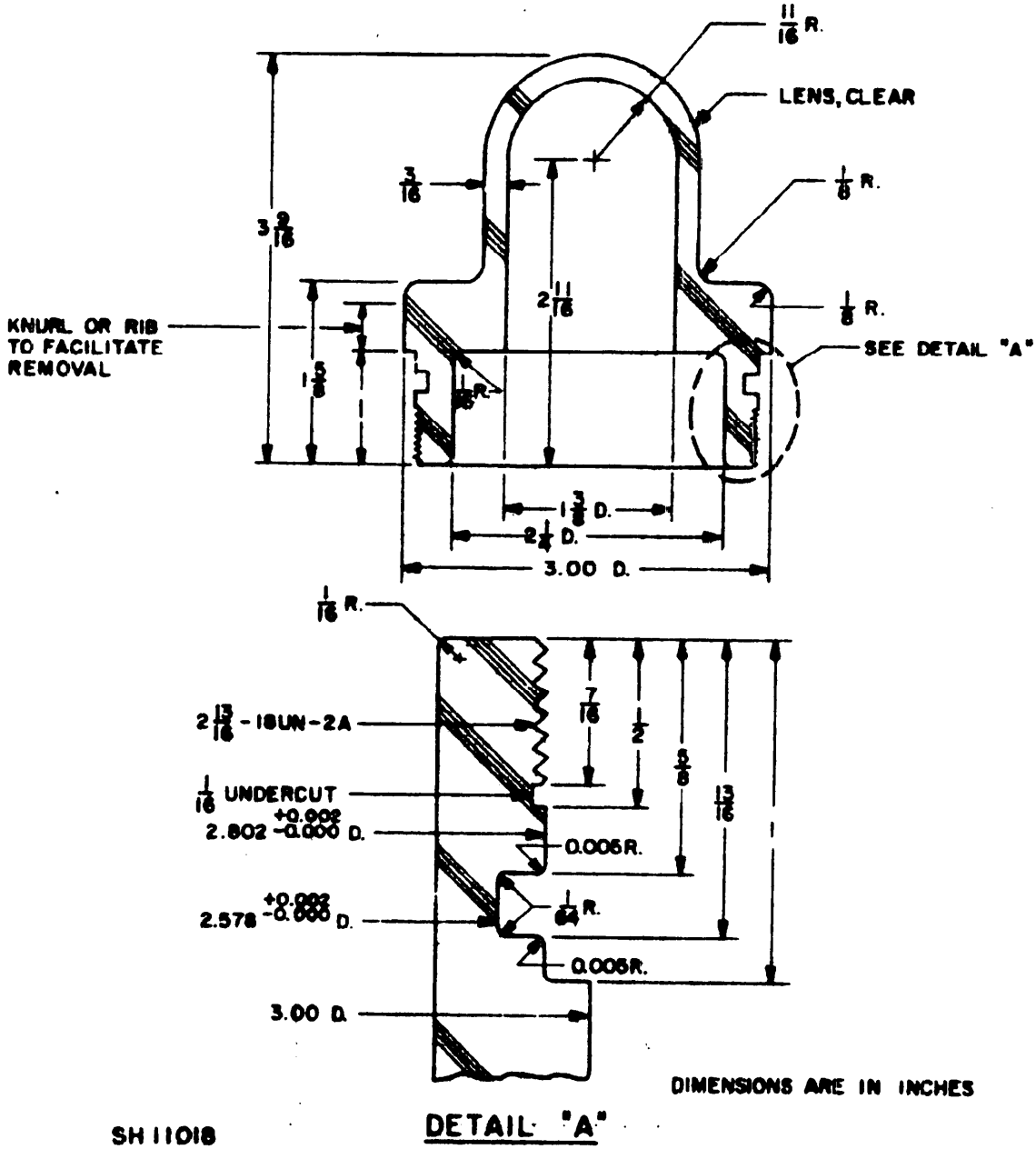


Figure 1 - Dimensions and design of lens.

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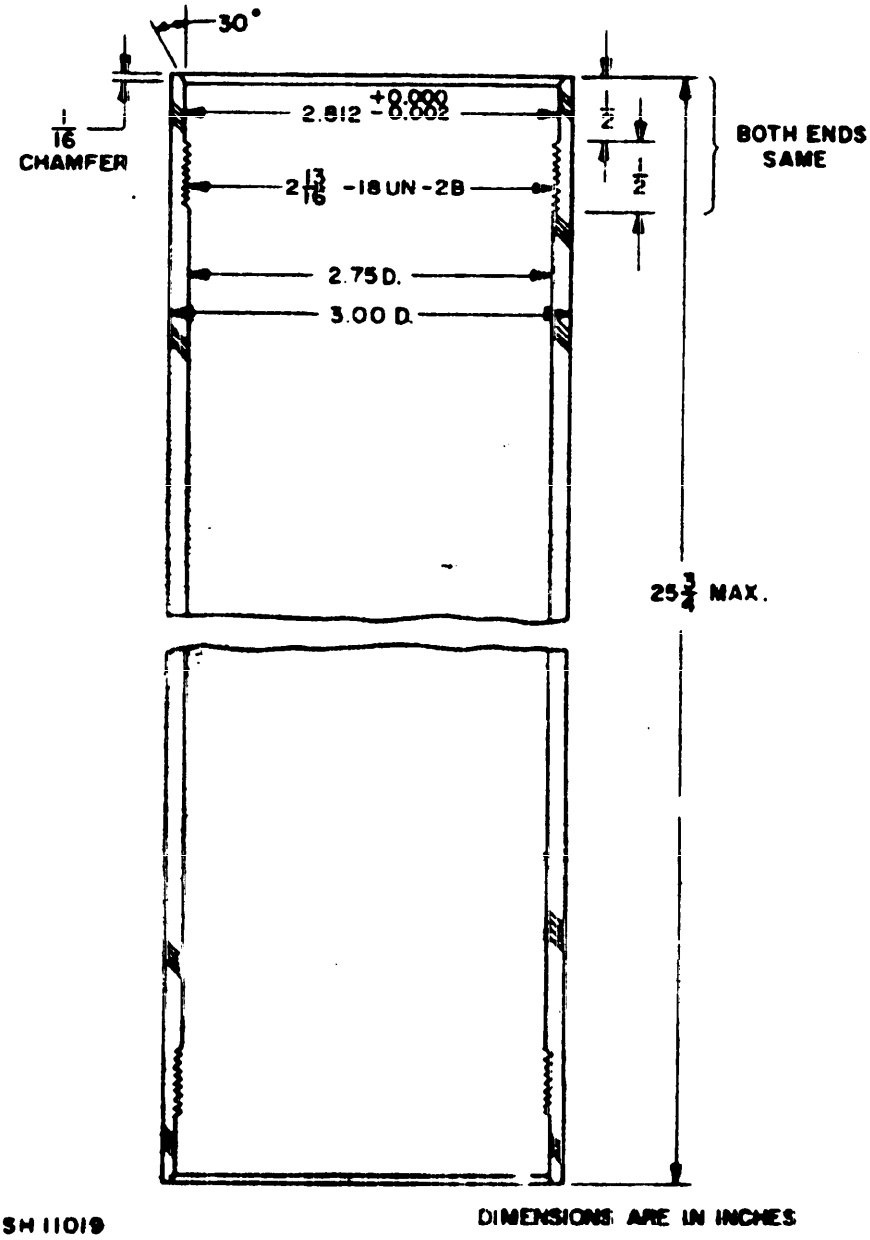


Figure 2 - Dimensions and design of case.

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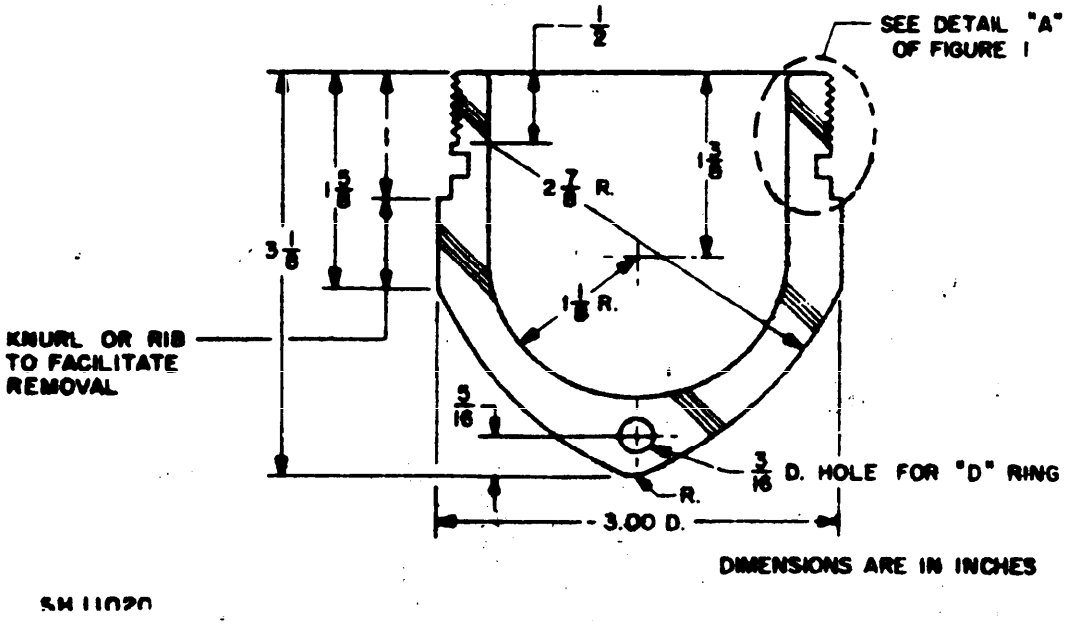


Figure 3 - Dimensions and design of base cone.

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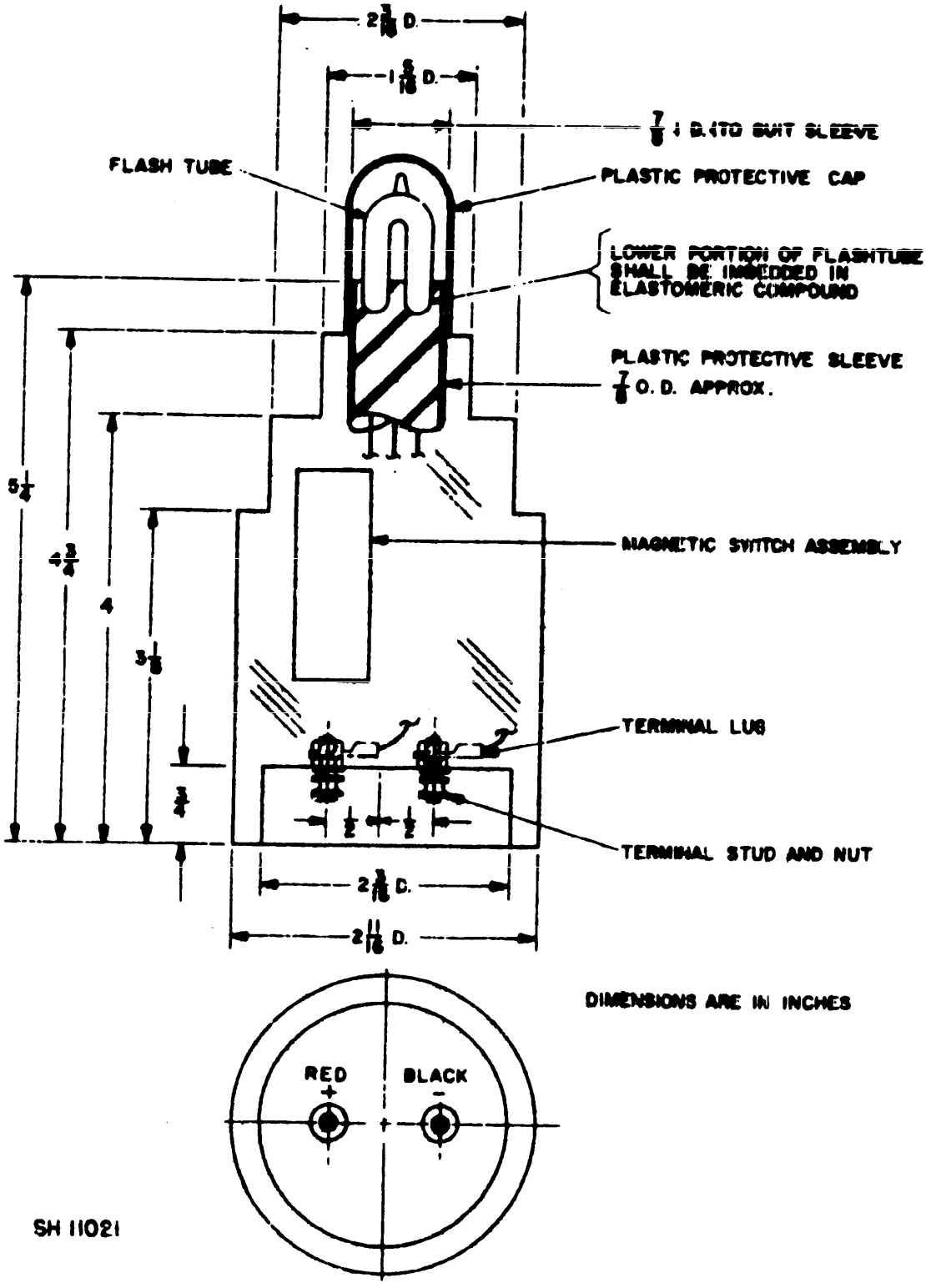


Figure 4 - Dimensions and design of electronic flashing circuit assembly.

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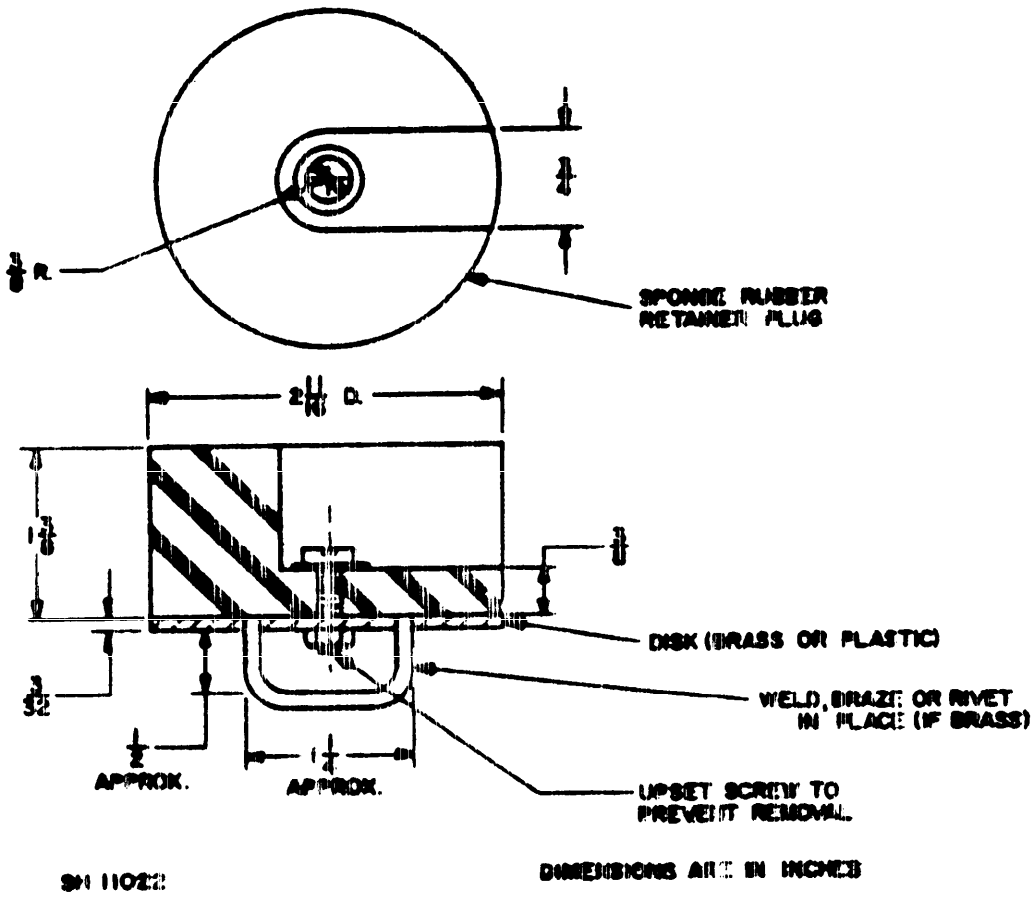


Figure 5 - Dimensions and design of battery retainer plug.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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